

4-cycle-free induced subgraphs of grid graphs

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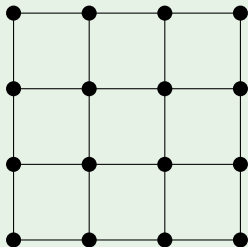
Grid Graphs

Definition

The *2-dimensional grid graph* has vertices $[n]^2 := \{1, \dots, n\}^2$, where the edges are those connecting (x, y) to (z, w) if and only if $|x - z| + |y - w| = 1$.

Grid Graphs

Example (4×4)



Induced Subgraphs

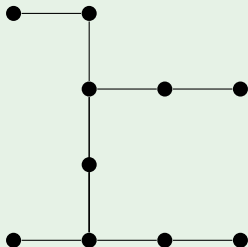
Definition

Suppose S is a subset of the vertices of a 2-dimensional grid graph G . Then the induced subgraph of G , denoted by G_S , is a subgraph of G with vertex set S , and uv is an edge in G_S if and only if uv is an edge in G and $u, v \in S$.

Induced Subgraphs

Example

If S comprises the vertices $(1, 1)$, $(2, 1)$, $(3, 1)$, $(4, 1)$, $(2, 2)$, $(2, 3)$, $(3, 3)$, $(4, 3)$, $(1, 4)$, and $(2, 4)$, then G_S will be the following:



Lemma 1.3

Lemma (Size of largest 4-cycle-free induced subgraphs)

Let n be even. Then the largest subset $S \subseteq V$ such that G_S is 4-cycle-free has size $3(n/2)^2$.

Lemma 1.3

Proof.

- Break the vertices in the $n \times n$ grid up into a disjoint union of 2×2 squares
 - ▶ Vertices of the form

$$\{(1 + 2k, 1 + 2\ell), (2 + 2k, 1 + 2\ell), (1 + 2k, 2 + 2\ell), (2 + 2k, 2 + 2\ell)\}$$



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- S can contain at most 3 of these vertices, for each such square, giving the upper bound $|S| \leq 3(n/2)^2$



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- S can contain at most 3 of these vertices, for each such square, giving the upper bound $|S| \leq 3(n/2)^2$
- If S just consists of all the 3 vertices

$$\{(1 + 2k, 1 + 2\ell), (2 + 2k, 1 + 2\ell), (1 + 2k, 2 + 2\ell)\},$$

then G_S is 4-cycle-free $\implies$ upper bound is attained



Main Theorem (1.4)

Theorem (Aiba and Croot, 2026+)

Let $\mathcal{F}$ denote the family of all subsets $S \subseteq V$ such that G_S is 4-cycle-free. For $n \geq 2$ and $\varepsilon \geq 0$, let

$$\mathcal{F}_\varepsilon := \{S \in \mathcal{F} : |S| \geq (3/4 - \varepsilon)n^2\}.$$

Then:

- (1) $|\mathcal{F}_0| \leq (n/2)^n$, if n is even; and we get the upper bound $((n+1)/2)^{n+1}$ if n is odd.
- (2) And for $\varepsilon > 0$ we have

$$(c - o(1))\varepsilon \ln(1/\varepsilon)n^2 \leq \ln |\mathcal{F}_\varepsilon| \leq \frac{(\varepsilon^{1/3}(\ln 1/\varepsilon)^{2/3} + f(\varepsilon))n^2}{3},$$

where $f(\varepsilon) = O(\varepsilon \ln(1/\varepsilon))$ and $c = \ln(4/3)$.

Literature Review

- Previous studies on largest induced *forests* (i.e., k -cycle-free for all $k \geq 3$)
- Alon *et al.* (2001): If G is a graph where the maximal degree of vertices is $\Delta \geq 3$, and if G has N vertices and independence number $\alpha(G)$, then the size of the largest induced forest in G has size at least

$$\alpha(G) + \frac{N - \alpha(G)}{(\Delta - 1)^2}$$

- For grid graphs $G^{(d)}$, $\alpha(G^{(d)})$ is $(1/2 + o(1))n^d$, so using $\Delta = 2d$, any largest induced forest S in $G^{(d)}$ satisfies

$$|S| \geq n^d \left(\frac{1}{2} + \frac{1}{2(2d-1)^2} + o(1) \right)$$

Literature Review

- Caragiannis *et al.* (2002): Found upper and lower bounds for the size of maximal induced forests for d -dimensional grid graphs to be $\frac{(d-1)n^d}{2d-1}$ and $\frac{(d-1)n^d - dn^{d-1} + 1}{2d-1}$, respectively
- This establishes the asymptotic order of the size of maximal induced forests in such graphs, which is $\frac{(d-1-o(1))n^d}{2d-1}$
- We (Aiba and Croot) originally researched maximal induced forests, but our results coincided with Caragiannis *et al.* (2002), notably the zig-zag construction for 2-dimensional grid graphs

Research Questions

Some problems worthy of consideration include:

- (a) What is the largest subset of S of $[n]^2$ such that H is 4-cycle-free?
- (b) How many largest 4-cycle-free induced subgraphs are there? That is, we are counting the number of induced 4-cycle-free subgraphs with the greatest number of vertices, in total.
- (c) How many maximal 4-cycle-free induced subgraphs are there? Here, a subset S of vertices is said to correspond to a **maximal**, 4-cycle-free induced subgraph if any addition of vertices to S results in an induced subgraph containing a 4-cycle. Note that that is a different concept from **largest** 4-cycle-free induced subgraph.
- (d) How many 4-cycle-free induced subgraphs H are there?

We answered (a) with Lemma 1.3 (Size of largest 4-cycle-free induced subgraphs).

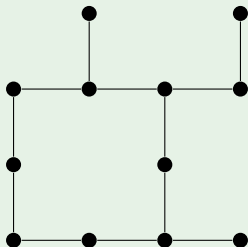
Letter Patterns

We associate to S , $|S| = 3n^2/4$, two different $m \times m$ arrays A and B of letters. Every position in the first array will have letters L and R, and every position in the second array will have letters U and D.

- If either $(2i + 1, 2j + 1)$ or $(2i + 1, 2j + 2)$ is missing from S , then the letter in the (i, j) position of A will be L.
- If either of $(2i + 2, 2j + 1)$ or $(2i + 2, 2j + 2)$ is missing from S , then the letter in the (i, j) position of A will be R.
- If either $(2i + 1, 2j + 1)$ or $(2i + 2, 2j + 1)$ is missing from S , then the letter in the (i, j) position of B will be D.
- If either of $(2i + 1, 2j + 2)$ or $(2i + 2, 2j + 2)$ is missing from S , then the letter in the (i, j) position of B will be U.

Letter Patterns

Example



The two arrays A and B that this corresponds to will be

$$\begin{array}{cc} L & L \\ R & R \end{array} \quad \text{and} \quad \begin{array}{cc} U & U \\ U & U \end{array} .$$

Necessary Condition

In order for G_S not to contain 4-cycles, we need that the arrays A and B avoid the patterns

$$\text{LR} \quad \text{and} \quad \begin{array}{c} \text{U} \\ \text{D} \end{array},$$

respectively.

Example

We could have 4-cycles such as

$$(2i+2, 2j+1) \rightarrow (2i+3, 2j+1) \rightarrow (2i+3, 2j+2) \rightarrow (2i+2, 2j+2) \rightarrow (2i+2, 2j+1).$$

These 4-cycles go in-between different 2×2 blocks.

Proof of Theorem 1.4 (1)

Proof.

- A forbids the pattern $LR \implies$ each row of A must be of the form

$LLL \cdots L$ or $RRR \cdots R$ or $RRR \cdots RLLL \cdots L$,

i.e., it must be a sequence of R 's followed by a sequence of L 's



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- There are at most m possible strings in each row $\implies$ there are at most m^m possible legal arrays A ; analogously, there are at most m^m possible arrays B



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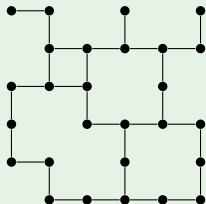
- There are at most m possible strings in each row $\implies$ there are at most m^m possible legal arrays A ; analogously, there are at most m^m possible arrays B
- There are most $m^m \cdot m^m = (n/2)^n$ possible pairs of legal arrays $A, B \implies$ there are at most this many different largest 4-cycle-free induced subgraphs



Condition is not sufficient

Example

L	L	L		D	U	U
R	R	R	and	D	U	U
L	L	L		D	U	U



This induced subgraph contains a 4-cycle with elements $(2,4)$, $(2,5)$, $(3,4)$, and $(3,5)$.

Maximal 4-cycle-free induced subgraphs

- Count the number of induced 4-cycle-free subgraphs having at least $(3/4 - \varepsilon)n^2$ vertices
- **Motivation:** When $\varepsilon > 0$ is small, these subgraphs are close to being the largest subgraphs we counted earlier
- Need to figure out how slightly lowering the lower bound on the number of vertices affects the count of the number of 4-cycle-free subgraphs

Upper Bound

- H is an induced subgraph of the $n \times n$ grid, H has at least $(3/4 - \varepsilon)n^2$ vertices
- Let

$$k = \left\lfloor \left(\frac{1}{12 \cdot \ln 2} \right)^{1/2} \left(\frac{\ln(1/\varepsilon)}{\varepsilon} \right)^{1/3} \right\rfloor,$$

and suppose n is divisible by $2k$

- **Idea:** Break $n \times n$ grid into $2k \times 2k$ subgrids with coordinates $(i, j) \in \{1, \dots, n/2k\}^2$, and count number of vertices in H in each subgrid
- Classify each subgrid by number of vertices from H (either $3k^2$ or $\leq 3k^2 - 1$)

Upper Bound

- Given $Z \subset \{1, \dots, n/2k\}^2$, $F_1(Z)$ = total number of possible subsets of all the vertices in all the subgrids (i, j) , where $(i, j) \in Z$
- $F_0(Z)$ = total number of possible subsets of all the vertices in those subgrids where each 2×2 subgrid of *those* contains exactly 3 vertices; so, each $2k \times 2k$ subgrid would contain $3k^2$ vertices
- Can show total number of induced subgraphs H that do not contain 4-cycles is at most

$$I := \sum_{\substack{Z_1 \subseteq \{1, \dots, n/2k\}^2 \\ |Z_1| \leq \varepsilon n^2}} F_1(Z_1) F_0(Z_0),$$

where Z_0 is the complement of Z_1

Upper Bound

From our choice on k , this gives

$$|I| \ll \exp\left(\frac{(\varepsilon^{1/3}(\ln 1/\varepsilon)^{2/3} + f(\varepsilon))n^2}{3}\right),$$

where $f(\varepsilon)$ has size $O(\varepsilon \ln(1/\varepsilon))$ and can be thought of as a “lower-order term”.

Lower Bound

- $S \subseteq V$ = any set of vertices where $|S| = 3n^2/4$, and G_S has no 4-cycles
- **Idea:** Count the number of subsets $S' \subseteq S$ with $|S'| \geq (3/4 - \varepsilon)n^2$
- Like the subgraph G_S , the induced subgraphs $G_{S'}$ are also 4-cycle-free

Lower Bound

For $\varepsilon \leq 3/8$, the number of such S' is at least

$$\sum_{k \leq \varepsilon n^2} \binom{3n^2/4}{k} \geq \binom{3n^2/4}{\lceil \varepsilon n^2 \rceil} \geq \left(\frac{3}{4\varepsilon} \right)^{\varepsilon n^2 - 1} \geq \exp(c\varepsilon \ln(1/\varepsilon)n^2),$$

where we can take $c = \ln(4/3)$.

Bibliography

- ① N. Alon, D. Mubayi, and R. Thomas. “Large induced forests in sparse graphs”. In: *Journal of Graph Theory* **38** (2001), 113–123.
- ② I. Caragiannis, C. Kaklamanis, and P. Kanellopoulos. “New bounds on the size of the minimum feedback vertex set in meshes and butterflies”. In: *Information Processing Letters* **83** (2002), 275–280.

Thank You!

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